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B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, JULY 2024
Fourth Semester

21CSE253T – INTERNET OF THINGS

(For the candidates admitted from the academic year 2022-2023 onwards)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

PART – A (20 × 1 = 20Marks)

Marks BL CO PO

Answer **ALL** Questions

- | | | | | |
|--|---|---|---|---|
| 1. Name the computing paradigm involves processing data closer to the data source rather relying on a central location. | 1 | 1 | 1 | 1 |
| (A) Cloud computing | | | | |
| (B) Edge computing | | | | |
| (C) Fog computing | | | | |
| (D) Distributed computing | | | | |
| 2. Identify the role of edge computing in IoT? | 1 | 2 | 1 | 1 |
| (A) A process of analyzing data at the source of generation, rather than sending it to a centralized location | | | | |
| (B) A process of analyzing data on a remote server | | | | |
| (C) A process of storing data on a local device | | | | |
| (D) A process of transmitting data over a wired network | | | | |
| 3. Name the wireless technology is known for its low power consumption and low range communication capabilities? | 1 | 1 | 1 | 1 |
| (A) Bluetooth | | | | |
| (B) Sigbee | | | | |
| (C) Wi-Fi | | | | |
| (D) LORAWAN | | | | |
| 4. Select the primary function of a gateway in IoT? | 1 | 2 | 1 | 1 |
| (A) Data processing | | | | |
| (B) Data storage | | | | |
| (C) Interfacing between different networks | | | | |
| (D) Data transmission | | | | |
| 5. NFC stands for _____. | 1 | 2 | 2 | 2 |
| (A) Near-field communication | | | | |
| (B) Network frequency control | | | | |
| (C) New form of connectivity | | | | |
| (D) National file consortium | | | | |
| 6. LR-WPANs standards from basis of specifications for high level communication protocol such as _____. | 1 | 2 | 2 | 2 |
| (A) Zigbee | | | | |
| (B) Allsean | | | | |
| (C) Tyrell | | | | |
| (D) Microsoft Azure | | | | |
| 7. The huge number of devices connected to the internet of things has to communicate automatically, not via humans. What is this called? | 1 | 2 | 2 | 2 |
| (A) Skynet | | | | |
| (B) Bot 2 bot | | | | |
| (C) Machine 2 machine | | | | |
| (D) Intercloud | | | | |

8. Which protocol is light weight? 1 4 2 2
 (A) HTTP (B) MQTT
 (C) COAP (D) SPI
9. How many types of Arduinos do we have? 1 5 3 1
 (A) 5 (B) 7
 (C) 6 (D) 8
10. _____ instruction set architecture is used in Raspberry Pi? 1 3 3 3
 (A) ARM (B) X86
 (C) AVR (D) MSP
11. Arduino IDE consist of 2 functions. What are they? 1 1 3 3
 (A) Loop () and build () (B) Build () and loop ()
 (C) Setup () and loop () (D) Setup () and build ()
12. Benefit of using NoSQL databases? 1 2 3 1
 (A) Strict data modeling (B) Limited scalability
 (C) Easy schema evolution (D) Limited data storage capacity
13. In unstructured data, information is usually embedded in 1 1 3 3
 (A) Tables (B) Rows
 (C) Text, images or other formats (D) Columns
14. Hadoop is a framework that works with a variety of related tools. Common cohorts include _____. 1 4 4 2
 (A) Map reduce, Hive and HBase (B) Map reduce, MySQL and Google Apps
 (C) Map reduce, Hummer and Iguana (D) Map reduce, Heron and Trumpet
15. What is the key difference between supervised and unsupervised learning? 1 1 1 4
 (A) Supervised learning requires labeled data, unsupervised learning does not (B) Supervised learning predicts labels, while unsupervised learning discovers patterns
 (C) Supervised learning is used for classification, while unsupervised learning is used for regression (D) Supervised learning is always more accurate than unsupervised learning
16. In order to improve their competitiveness and services assurance, the _____ require independently funded IoT projects. 1 1 5 2
 (A) eGovernment related (B) Enterprise-based
 (C) Company based (D) Business oriented platform
17. An IoT _____ center is envisaged as an important part of the generic IoT platform to unify the organization. 1 3 5 4
 (A) Individual information (B) Individual integration
 (C) Integrated information (D) Individual and integrated information

18. The core element of architecture of smart city is _____.	1	3	5	2
(A) Urban application platform				
(B) Intergraded information provider				
(C) Management center				
(D) Mobile unified service				
19. IoT promotes the creation of IoT terminal industry _____.	1	1	5	4
(A) Devices				
(B) Network				
(C) Clusters				
(D) Things				
20. LOWPAN adaption layer contains?	1	2	5	4
(A) Header compression				
(B) Header compression, fragmentation and layer 2 forwarding				
(C) Fragmentation				
(D) Layer 2 forwarding				

PART – B (5 × 8 = 40 Marks)

Marks BL CO PO

Answer ALL Questions

21. a. Explain the packet structure and message format of MQTT protocol with a neat sketches.	8	2	1	1
(OR)				
b. Explain the enabling technologies behind IoT. Also, draw the architecture of IoT based applications.	8	1	1	1
22. a. Write about the key aspects of LORAWAN in content of IoT.	8	2	2	2
(OR)				
b. Mention about the limitations of IPv4 and the benefits of IPv6.	8	1	2	1
23. a.i. Compare the key highlights of IEEE 802.115.4g standard IEEE 802.15.4e standard.	4	2	2	1
ii. Write about the characteristics of constrained nodes.	4	2	2	2
(OR)				
b. Explain the procedure to build IoT with Raspberry Pi. What are the physical devices and end points?	8	1	3	3
24. a. Generalize in detail about Apache spark and Apache Kafka with data flow diagram.	8	6	4	2
(OR)				
b. Evaluate the working of Xively cloud dashboard devices for IoT by giving suitable necessary explanation.	8	5	4	5
25. a. Formulate an Industrial application of IoT system and brief on the various use case of smart and connected cities.	8	6	5	4
(OR)				
b. Explain in detail about Cisco Grid blocks reference model with architecture and benefits.	8	2	5	2

PART – C (1 × 15 = 15 Marks)

Answer ANY ONE Question

	Marks	BL	CO	PO
26. Generalize the purpose of python web application framework – Django and Amazon web service for IoT.	15	6	4	2
27.i. Discuss active and passive sensor.	2	2	2	1
ii. Compare the key features of COAP and MQTT.	6	2	2	1
iii. Illustrate with a neat sketch IoT design for smart agriculture system and justify now privacy and security provided in your prototype.	7	3	5	4

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